

KROPPEFJALL-GRANAN, SwedenWMO: **025400**Lat: **58.6065N**Lon: **12.1973E**Elev: **571**StdP: **14.40**Time Zone: **1.00 (EUC)**Period: **01-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	8.2	12.3	0.3	5.7	14.1	6.3	7.8	14.0	15.7	40.4	14.2	39.3	3.0	60	0.557

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.4	78.0	64.0	74.9	62.6	71.6	61.0	66.2	74.1	64.5	72.1	62.7	69.0	4.1	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.6	89.9	68.3	61.8	84.2	66.8	60.1	79.3	65.2	31.2	74.4	29.9	72.2	28.6	69.2	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.9	12.1	10.6	4.9	82.4	6.3	2.6	0.4	84.3	-3.4	85.8	-6.9	87.2	-11.5	89.1		
(5)			4.5	69.1	6.3	2.1	0.0	70.6	-3.7	71.8	-7.3	72.9	-11.8	74.4	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.2	28.6	28.6	33.8	42.4	50.8	57.6	61.7	59.8	53.0	44.5	36.9	32.1	(6)	
	DBStd	13.32	7.73	7.61	6.97	5.87	6.47	5.13	4.87	4.16	4.97	5.99	6.65	8.35	(7)	
	HDD50	3235	664	600	502	238	66	3	0	0	27	187	392	556	(8)	
	HDD65	7621	1129	1020	967	677	441	230	129	169	360	636	842	1021	(9)	
	CDD50	1132	0	0	0	11	91	231	363	303	117	16	0	0	(10)	
	CDD65	44	0	0	0	0	1	8	27	7	1	0	0	0	(11)	
	CDH74	356	0	0	0	0	11	87	218	38	2	0	0	0	(12)	
	CDH80	35	0	0	0	0	0	7	27	1	0	0	0	0	(13)	
	WSAvg	4.7	5.1	5.0	5.3	4.9	4.6	4.5	4.2	3.7	4.3	4.2	4.9	5.3	(14)	
(15) Precipitation	PrecAvg	37.4	3.1	2.4	2.0	2.2	2.5	3.3	3.6	3.4	3.4	4.1	3.7	3.5	(15)	
	PrecMax	50.4	6.3	5.3	4.6	5.0	3.3	5.4	8.0	7.0	7.6	8.9	6.9	7.4	(16)	
	PrecMin	26.7	0.5	0.7	0.7	0.4	0.9	1.8	0.8	1.0	1.0	1.3	1.0	0.6	(17)	
	PrecStd	6.5	1.7	1.3	1.0	1.5	0.8	1.2	1.7	1.6	1.8	2.0	1.7	1.9	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.9	46.9	56.0	66.0	75.6	80.9	83.4	77.8	71.9	60.4	52.6	48.7	(19)	
		MCWB	42.0	40.6	46.4	51.2	58.4	62.7	65.7	64.7	62.4	55.5	51.2	47.6	(20)	
	2%	DB	42.3	42.2	49.9	61.4	71.2	76.3	79.2	74.5	66.4	56.6	50.2	46.5	(21)	
		MCWB	40.2	39.5	42.4	49.8	56.6	61.5	65.0	64.0	59.0	53.6	48.8	45.6	(22)	
	5%	DB	40.6	40.3	45.5	56.6	67.0	72.1	76.3	71.1	63.0	54.7	48.4	44.7	(23)	
		MCWB	38.2	38.4	39.8	46.8	55.4	59.6	64.2	62.1	57.5	52.6	47.2	43.1	(24)	
	10%	DB	38.9	38.6	42.6	52.5	62.2	68.3	72.3	67.7	60.7	52.9	46.2	42.8	(25)	
		MCWB	37.0	37.0	38.8	44.6	53.3	58.0	62.6	60.3	56.0	51.0	45.1	41.3	(26)	
	0.4%	WB	43.0	42.3	46.8	53.7	62.2	65.8	69.3	68.0	63.8	56.7	52.1	48.0	(27)	
		MCDB	44.1	44.0	53.9	63.7	71.3	74.1	77.0	73.2	69.3	58.3	52.3	48.2	(28)	
(29) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	40.8	40.3	43.7	50.6	59.2	63.1	67.1	65.2	60.4	54.6	49.2	45.8	(29)	
		MCDB	42.1	41.3	48.2	58.8	65.8	73.0	75.6	71.5	64.3	56.0	49.7	46.4	(30)	
	5%	WB	39.0	38.7	41.2	48.2	56.7	61.1	65.4	63.4	58.6	52.9	47.3	43.3	(31)	
		MCDB	40.2	40.0	44.6	55.1	63.5	69.7	73.7	68.9	62.1	54.3	48.1	44.2	(32)	
	10%	WB	37.3	36.9	39.4	45.8	54.3	59.0	63.8	61.7	56.9	51.4	45.3	41.2	(33)	
		MCDB	38.6	38.5	41.7	51.0	60.5	66.0	70.9	66.1	59.8	52.7	46.2	42.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.4	7.8	11.2	15.0	15.9	15.9	15.4	13.9	12.7	9.8	7.2	7.1	(35)	
		MCDBR	8.7	9.1	15.7	21.6	21.5	21.9	21.9	18.8	15.8	10.7	8.9	8.0	(36)	
		MCWBR	8.7	7.1	10.0	12.5	11.6	11.4	10.6	10.1	9.4	8.2	8.6	8.1	(37)	
	5% WB	MCDBR	8.8	8.2	14.0	18.9	18.1	19.2	19.2	16.2	14.4	9.9	8.5	8.1	(38)	
		MCWBR	8.8	6.8	9.7	11.6	10.7	11.1	10.3	9.6	9.4	8.3	8.4	8.4	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.279	0.312	0.332	0.369	0.365	0.354	0.375	0.371	0.341	0.330	0.298	0.260	(40)	
	taud		2.235	2.317	2.375	2.330	2.408	2.460	2.419	2.427	2.499	2.438	2.299	2.162	(41)	
	Ebn at Noon		175	227	258	265	275	280	270	263	254	219	166	138	(42)	
	Edh at Noon		14	22	27	34	33	32	33	30	24	19	14	11	(43)	
(44) All-Sky Solar Radiation	RadAvg		106	285	767	1244	1653	1868	1678	1319	888	398	139	72	(44)	
	RadStd		19	50	74	158	143	102	198	107	101	62	28	7	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (62 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page